

Individual Project Report

Campus360: Mobile-Based Campus Navigation System

Name: Sha, Shaik Ibtihajulla

Email: ibsh25@student.bth.se

Course: Mobile Application Development

Institution: Blekinge Institute of Technology

Submission Date: June 22, 2026

1 About Me

1.1 Previous Experience with Group Programming Assignments

Before Campus360, I had prior experience working on group programming assignments where the primary focus was completing functional requirements. In those projects, collaboration was often limited to splitting tasks, with minimal discussion about architecture or long-term maintainability.

This project differed significantly, as it required closer coordination and a shared understanding of architectural decisions. Each change could affect other parts of the application, which made collaboration more structured and deliberate.

1.2 Personal Learning Strategy for Kotlin and Android Development

My learning strategy was incremental and practice-oriented. I focused first on understanding Kotlin fundamentals such as null safety, immutability, and data classes before applying them in an Android context.

Once comfortable with Kotlin, I concentrated on Android-specific concepts like lifecycle management, ViewModels, and Jetpack Compose. Rather than memorizing APIs, I tried to understand why certain patterns were recommended and how they improved code reliability.

1.3 Preferred Learning Resources

My primary learning resource was the official Android developer documentation. I frequently referred to documentation related to Jetpack Compose, navigation, and recommended architecture patterns.

In addition, examining existing project code and experimenting with small changes helped reinforce theoretical knowledge through practical observation.

1.4 Team Collaboration

Collaboration within the team involved regular communication about design choices and implementation progress. We discussed navigation flow, screen responsibilities, and UI consistency before integrating changes.

Version control played an important role in collaboration. Reviewing commits and resolving merge conflicts helped improve my understanding of collaborative development practices.

1.5 Task Allocation

Task allocation was based on individual interests and learning goals. Responsibilities were divided so that each member could contribute meaningfully while gaining experience in specific areas of mobile development.

2 Application Design and Development Process

2.1 Why Campus360 Was Developed as a Mobile Application

Campus360 was developed as a mobile application because campus navigation is inherently a mobile use case. Users typically move between buildings while relying on their smartphones for guidance and information.

Mobile devices provide portability, instant access, and intuitive touch-based interaction, which are essential for navigation scenarios. A desktop or web application would be less practical in real-time campus movement contexts.

2.2 Development Process and Comparison to Traditional Methods

The development process followed an incremental approach, where features were added and refined through multiple iterations. Unlike traditional web or desktop development, Android development required careful handling of lifecycle events and state preservation.

Jetpack Compose introduced a reactive UI paradigm, where the interface responds automatically to state changes. This differed from imperative UI models commonly used in desktop or web applications.

2.3 Influence of UI Design and Prototyping on Development

UI design and prototyping influenced later development by clarifying navigation flow and screen responsibilities early in the process. Having a defined UI structure reduced uncertainty during implementation and helped maintain consistency across screens.

Early design decisions also simplified state management, as ViewModels could be aligned with specific screens and user interactions.

3 Security Risks and MASVS Testing

3.1 Security Risks Specific to Campus360

Risk 1: Integrity of Locally Stored Navigation Data

One security risk in Campus360 relates to the local storage of Points of Interest (POI) data. Since the application stores POI information locally to support offline usage, there is a risk that this data could be modified or corrupted. From a user perspective, altered navigation data could lead to incorrect directions, causing confusion or inconvenience, particularly for new students or visitors.

This risk matters because users trust the application to provide accurate navigation information. Mitigation could include validating locally stored data, controlling update mechanisms, or restricting access to internal storage structures.

Risk 2: Lack of Transparency Regarding Data Handling

A second security risk involves the lack of transparency regarding data handling. While Campus360 does not request location permissions or collect personal data, users may still assume that navigation applications track their movements. This mismatch between user perception and actual functionality could lead to misunderstandings or misuse of the app.

Clear communication within the app about what data is and is not collected would help mitigate this concern and improve user trust.

3.2 OWASP MASVS Assessment

MASVS-STORAGE: Secure Local Data Storage

The MASVS storage requirement states that applications should protect locally stored data from unauthorized access or modification.

In Campus360, POI data is stored locally to enable offline navigation. While this data is not highly sensitive, improper handling could still impact data integrity and user experience.

Campus360 partially complies with this requirement. The application avoids storing personal user information; however, no encryption or integrity verification mechanism is implemented for locally stored POI data.

Testing was performed through manual code review and emulator-based testing. The data storage approach was inspected, and no encryption libraries or secure storage mechanisms were identified.

MASVS-PLATFORM: Secure Use of Platform Features

This MASVS requirement emphasizes secure usage of platform features and minimizing exposure through debugging or system interfaces.

Campus360 was tested exclusively on an Android emulator, and no debug logging was observed in the application code. This reduces the risk of leaking internal application behavior

through system logs.

Campus360 largely complies with this requirement. The absence of logging statements such as `Log.d` reduces exposure risks; however, additional platform-level protections could be considered for future releases.

Testing involved emulator-based execution and monitoring application behavior during navigation and screen transitions.

4 Ethical Concerns

4.1 Ethical or Moral Concern Related to Campus360

An ethical concern related to Campus360 is user reliance on navigation accuracy. Users may assume that the application always provides correct and up-to-date information. If navigation data becomes outdated, users could be misled without realizing the limitation.

Developers have an ethical responsibility to ensure accuracy or clearly communicate potential limitations to users.

4.2 Dark Patterns in Campus360

One potential dark pattern in Campus360 is implicit trust reinforcement through UI design. The application presents navigation information confidently without indicating uncertainty or data freshness.

Although unintentional, this could lead users to over-rely on the app. Introducing subtle indicators such as update timestamps or disclaimers could help users make more informed decisions.

Appendix

The following files represent the components of the Campus360 application that I was primarily responsible for during the development process.

- **Main application entry point**

- MainActivity.kt

- **Data models**

- POIModels.kt

- **ViewModel**

- HomrViewModel.kt

- **User interface screens**

- POIScreen.kt

- POIDetailsScreen.kt

- NotificationsScreen.kt

References

- [1] OWASP, *Mobile Application Security Verification Standard*. Available: <https://mas.owasp.org>
- [2] OWASP, *Mobile Application Security Testing Guide*. Available: <https://mas.owasp.org/MASTG/>
- [3] Android Developers, *Android Documentation*. Available at: <https://developer.android.com>
- [4] Android Developers, *Jetpack Compose Documentation*. Available at: <https://developer.android.com/jetpack/compose>
- [5] Google, *Material Design 3*. Available at: <https://m3.material.io>